

Final Project

In this project, you will work with a real-world dataset, perform ETL (Extract, Transform, Load) processes, build a data warehouse or data cube, and create insightful visualizations. You are required to use these tools: **SSAS (SQL Server Analysis Services), Power BI, SQL Server Management Studio (SSMS), Alteryx, Apache Spark, or Snowflake.**

This project is in two parts:

1. The first part of this project is to select a company business and demonstrate the use of data by applying the different topics that we have used to date to show a business result – this could be in the form of a report or a series of graphical representations that gives insights into the business activities of that organization
2. The second part of this project is to prepare a power point slide presentation where you will be given a 10-15 minutes window to present to the class the business and what it does and what your analysis found by explaining the results and why you used the different technologies in the process.

Project Details

Using 5 stages in big data analysis, come up with a scenario to solve a particular problem with big data.



Step 1: Data Selection & Exploration

- Choose a dataset from a public repository (e.g., Kaggle, Data.gov, Google Dataset Search) or an enterprise data source. (Used more than 3 data sources including at least 100k records)
- Describe the dataset (size, structure, data types, missing values, key attributes).
- Use **SQL Server Management Studio (SSMS)** to explore and clean the dataset.
- Identify a business question or problem that your analysis will address.

Deliverable: A short report describing the dataset and business problem.

Step 2: Data Preprocessing & ETL

- Use **Alteryx, Apache Spark, or Snowflake** to clean, transform, and preprocess the data.
- Handle missing data, duplicate values, and inconsistent records.
- Create necessary joins, aggregations, and calculated fields.

Deliverable: A process flow diagram and a brief explanation of the ETL steps taken.

Step 3: SSAS Tabular Model

- Create dimension and fact tables in **SQL Server** or **Snowflake**.
- Use **SSAS** to build a tabular model
 - Define measures.
 - Implement aggregations and calculated members.
- Test the cube with sample queries.

Deliverable: ERD diagram, SQL script for schema creation, and screenshots of the model.

Step 4: Data Analysis & Queries

- Write **at least five SQL queries** to analyze the data (e.g., trend analysis, ranking, customer segmentation).
- Run queries in **SSMS, Apache Spark (PySpark/SQL), or Snowflake**.

Deliverable: Query scripts and their results.

Step 5: Visualization & Reporting in Power BI

- Connect Power BI to **SSAS or SSMS**.
- Create **at least five visualizations**.

Deliverable: A Power BI file (.pbix) with the final explanation.

Step 6: Final Presentation & Documentation

- **Presentation:** A 10-15 minute presentation summarizing:
 - The problem statement
 - The ETL process
 - Data model
 - Key insights from Power BI
- **Documentation:** A final report containing:
 - Introduction & dataset description
 - ETL methodology
 - Schema and SSAS tabular model
 - Key queries & insights
 - Challenges and solutions

Deliverable: PowerPoint presentation & project report.

Distribution of Marks- Project

Data Collection	10
Data Transformation / Processing / Combining /ETL	20
Data visualization	10
Presentation on finding	10
Total	50